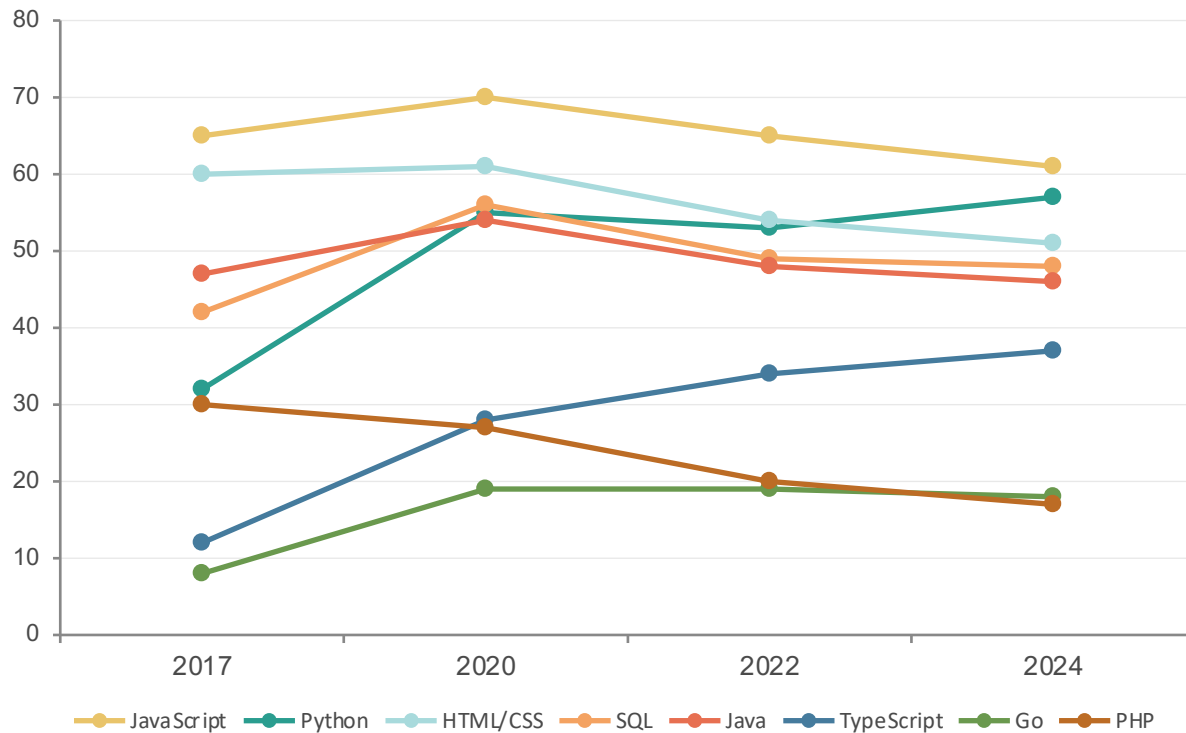




State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers on programming language usage, satisfaction, adoption intent, and domain specialization trends from 2017 to 2024.

Python Closes the Gap: 8 Years of Language Usage Shifts



Python

32% → 57%

Nearly closed the gap with JavaScript, driven by AI/ML boom

TypeScript

12% → 37%

Tripled its share in 7 years; strongest sustained growth

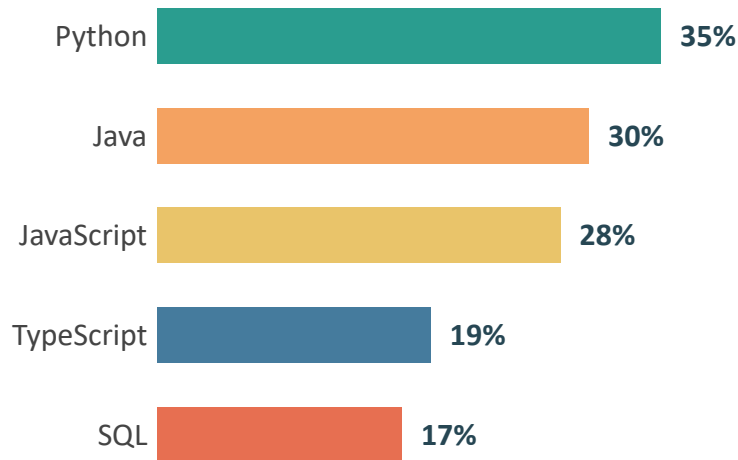
PHP

30% → 17%

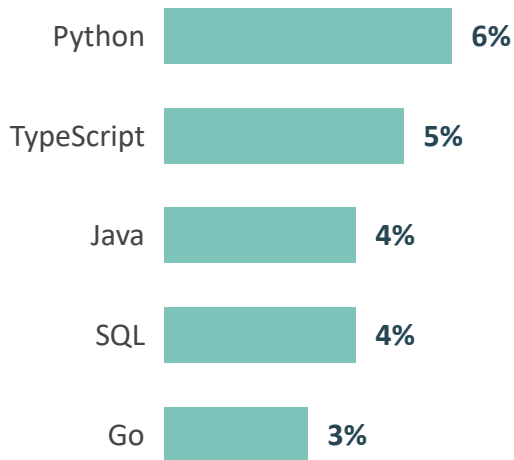
Significant decline as newer alternatives gain ground

Python Is Now the #1 Primary Language at 35%

Primary Language

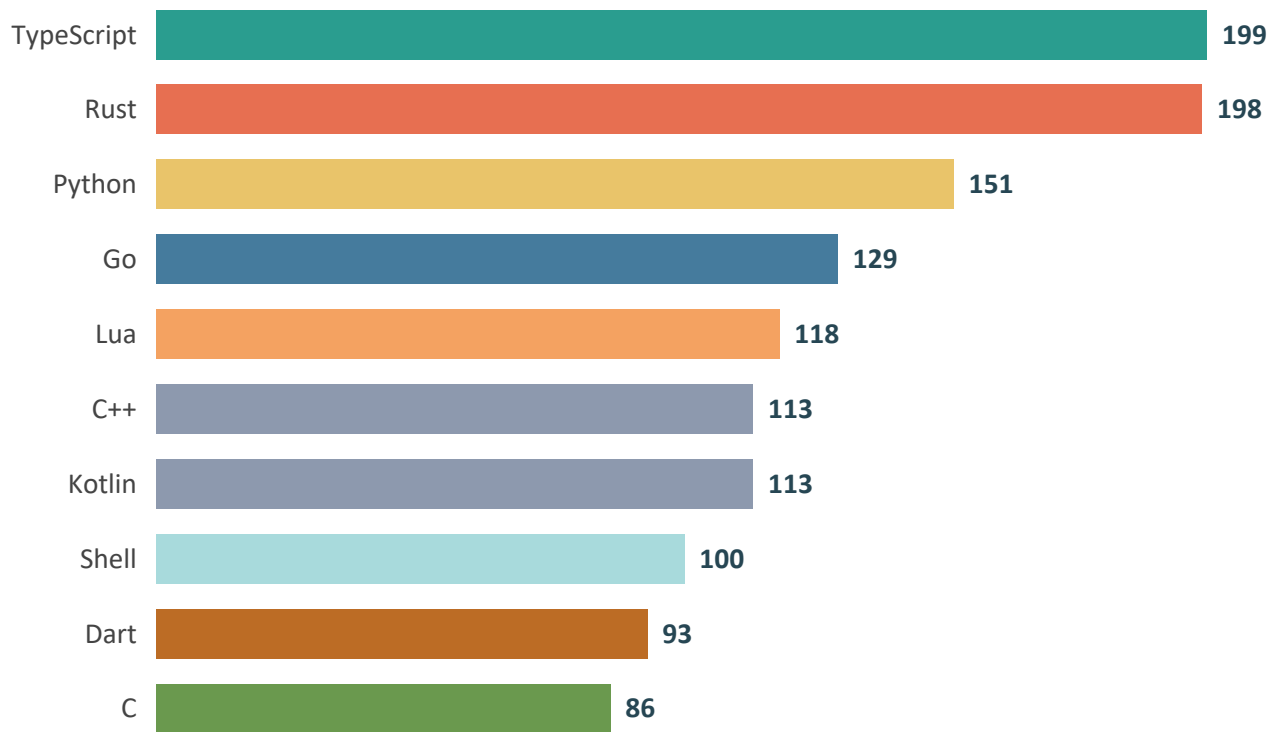


Adoption Intent



Key Insight: Developers stick with what they know. Adoption intent remains below 6% across all languages, suggesting strong ecosystem lock-in. Expertise investment and tooling familiarity keep developers loyal to their chosen stacks.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



What drives satisfaction?

- Type safety and expressive type systems
- Modern tooling and developer experience
- Strong compiler guarantees that reduce bugs
- Active community and ecosystem growth

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain	1st	2nd	3rd
AI / Machine Learning	Python 31%	Java 8%	C++ 5%
Data Processing & Analytics	Python 17%	SQL 15%	Java 10%
User Interface	TypeScript 62%	JavaScript 58%	Dart 8%
APIs & Services	Java 49%	Python 25%	TypeScript 18%
Application Logic & Workflows	Java 55%	Python 20%	C++ 10%
System Tools & Components	C++ 27%	Python 15%	Go 12%
Integrating with APIs & Services	Python 38%	JavaScript 25%	Java 15%

Source: JetBrains State of Developer Ecosystem Report 2024 (n=23,262)

What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML teams

Python is the undisputed leader in AI (31%), data science (28%), and data processing. It is now the #1 primary language at 35%.

02

TypeScript is the future of frontend and full-stack

Tripled from 12% to 37% usage. Leads developer satisfaction (score: 199) and dominates UI development at 62%.

03

Rust and Go are your infrastructure growth plays

Both prioritize performance and compiler safety. Go stable at 18%; Rust climbing to 11%. Ideal for cloud-native and systems work.

04

PHP and Ruby hiring pools are shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Plan migration paths and expect tighter talent markets.

05

Factor developer satisfaction into retention strategy

Satisfaction correlates with type safety and modern tooling. TypeScript (199) and Rust (198) top the charts for a reason.